Learning Task 2 - Flashing over soldering

The purpose of this Learning Task is to show how you can go from the analog to the digital domain by sampling and which precautions you need to take. The digital domain also has an concept of frequency, captured by the DFT. You will implement a RC lowpass analog filter as a digital filter, hence flashing an digital implementation on a development board (Figure 1 left) over soldering a resistor and a capacitor (Figure 1 right). We will compare the two implementations.

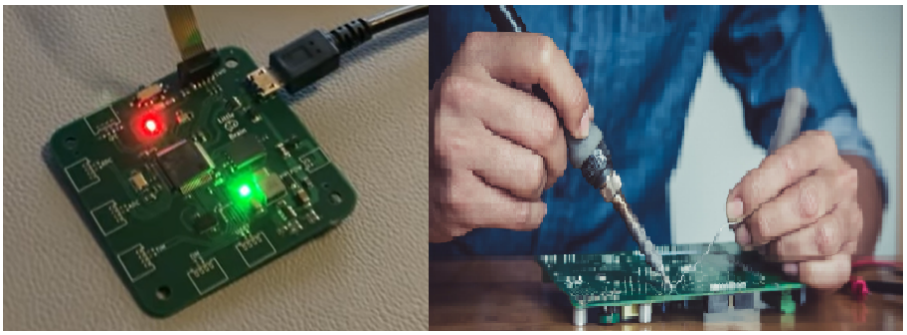


Figure 1: No soldering, just a digital implementation

Q1 Suppose we sample an analog signal $x_1(t) = \cos(200\pi t)$. We sample every $T = 10^{-3}s$. Give a formula for $x_1[n]$.
H1 What is the relation between $x_1(t)$ and $x_1[n]$?

Q2 Suppose we sample another analog signal $x_2(t) = \cos(2\pi f t)$ with an unknown frequency f. It results in the digital signal $x_2[n] = \cos(\frac{\pi}{2}n)$. We sample every $T = 10^{-3}s$, so $f_s = 1000Hz$. Find **all** frequencies f that lead to this $x_2[n]$. Can you generalize what you found?
H2 Periodicity?

Q3 Suppose now $x_3[n] = \cos(\Omega n)$ is a finite length signal of length 1024. We know it is periodic. Use autocorrelation to determine its (discrete) period. Also comment on the shape of the autocorrelation.
H3 Look for xcorr.

Q4 Determine the DFT $X_3[k]$ of $x_3[n]$ of the previous question and plot $|X_3[k]|$. At which index or indices would you expect peaks? Suppose now that $x_4[n]$ is sampled from the same analog signal $x_3(t)$, but now has length 2048 instead. Where would you expect a peak or peaks? And what about $x_5[k]$ with length 768?
H4 Use fft as implementation of DFT.

Q5 If $y[n] = x^*[n]$, what is the relationship between $X[k]$ and $Y[k]$? What happens if $x[n]$ is real?
H5 No hint

Q6 Suppose we want to implement the RC lowpass filter

$$V_C'(t) + \frac{1}{RC}v_C(t) = \frac{1}{RC}V_{in}(t).$$

digitally. This is a two-step process. First approximate the derivative by a difference

$$V_C'(t) \approx \frac{V_C(t) - V_C(t - \Delta t)}{\Delta t}$$

and then use $x[n] = V_{in}(nT),\ y[n] = V_C(nT)$. Derive the digital filter.
H6 What is Δt?

Q7 In the next three question we want to compare the effect of the analog filter with that of the digital filter with $f_s = 10^4$Hz. It can be shown that if $x(t) = A\cos(\omega_0 t)$ is fed to a linear filter with transfer function $H(\omega)$, then the filter output is given by

$$y(t) = A|H(\omega_0)|\cos(\omega_0 t + \arg(H(\omega_0)).$$

Note that $H(\omega)$ is complex valued in general, and the polar notation $H(\omega) = |H(\omega)|e^{j\arg(H(\omega))}$ helps to construct the output $y(t)$. The transfer function alters the amplitude and the phase of the input, but leaves the frequency intact. In our RC example, the transfer function is given by $H(\omega) = \frac{1}{1+j\omega RC}$. Suppose we pick $RC = \frac{1}{3675}$ and $\omega = 3675$, find $x(t)$ and $y(t)$ and plot them for $t \in [0, 0.025]$.
H7 $|z|^2 = zz^*$ and $\arg(\frac{1}{z}) = -\arg(z)$

Q8 Implement and apply the the digital filter on a sampled input from $t = 0$ to $t = 0.025$. Use $RC = \frac{1}{3675}$, $\omega = 3675$ and $f_s = 10^4$Hz.
H8 filter

Q9 Compare the analog and digital filter output and discuss observations.
H9 No hint